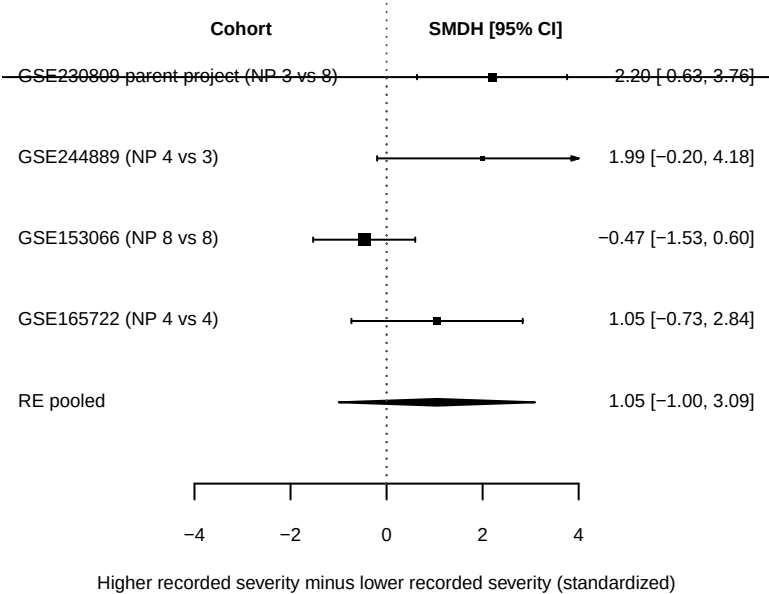
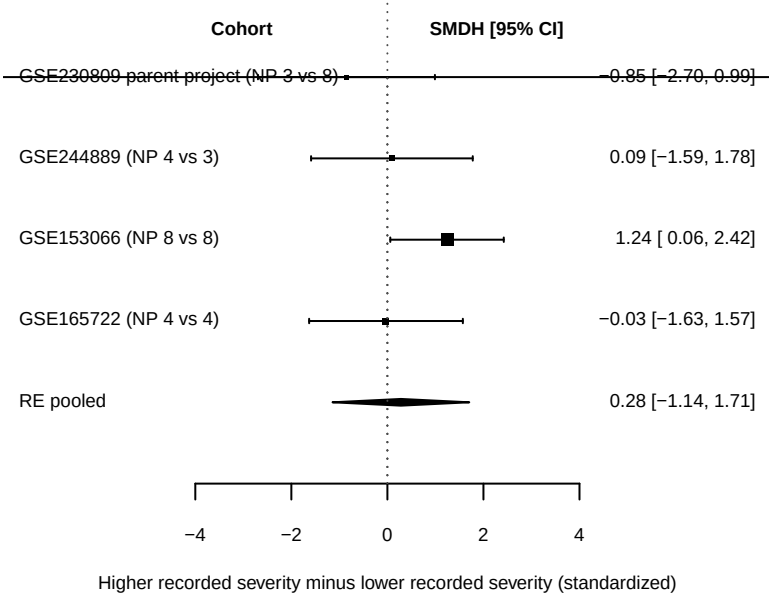


Supplementary Figure S3. Exploratory cohort-level random-effects meta-analysis of NP module scores.

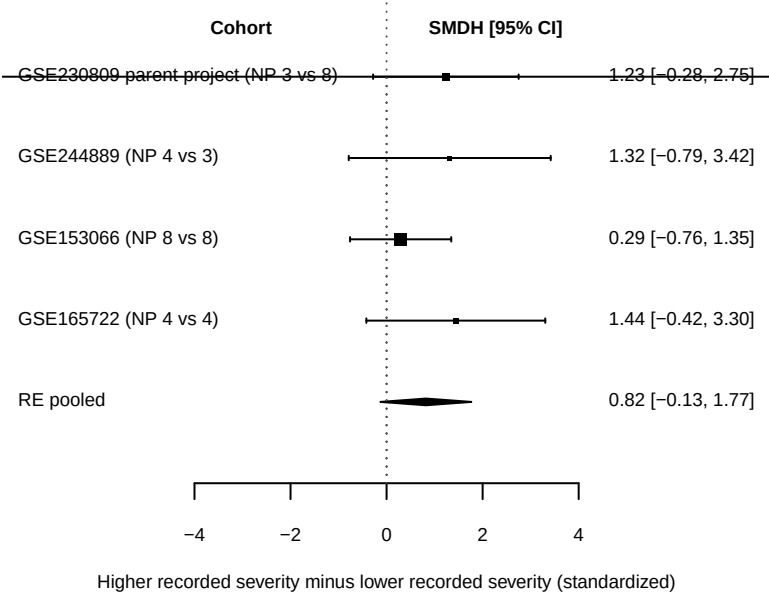
ECM / collagen remodeling



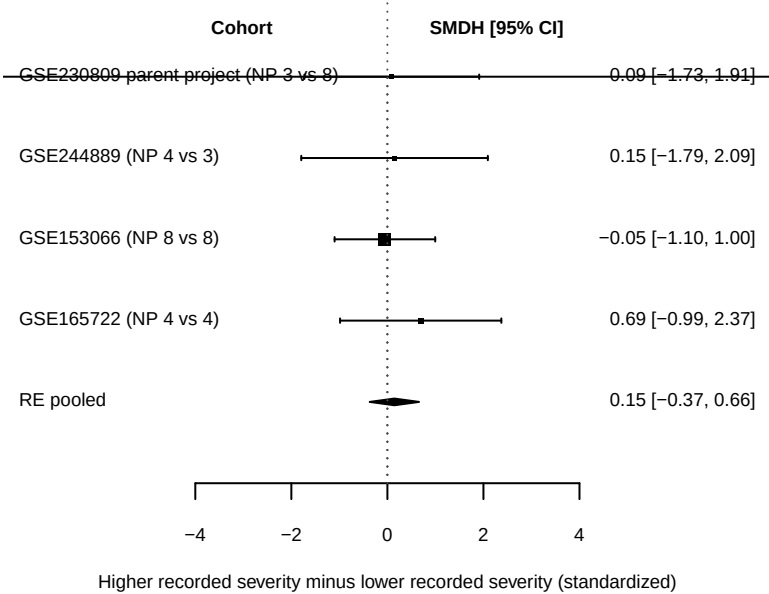
Inflammatory / NF- κ B



Hypoxia / oxidative stress



Disc matrix homeostasis



All cohorts remain non-confirmatory; pooling does not remove age, source, platform, or presumed-key limitations.

Primary standardized effect: SMDH; REML τ^2 ; Knapp-Hartung intervals; $k = 4$ per module.